

3D Localizers for Surgical Navigation 600.455/655

Russell H. Taylor

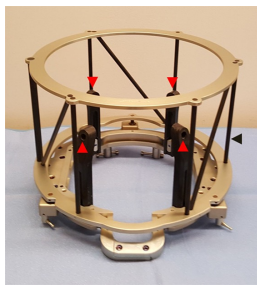
600. 455/655 Fall 2024 Copyright R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology

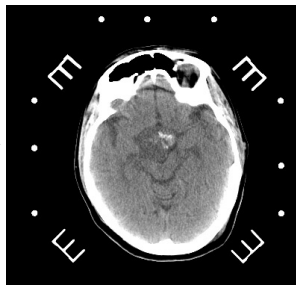


1

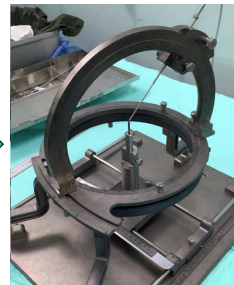
Traditional Stereotactic Systems



**Cosman-Roberts-Wells
(CRW) head frame**



CT Image



**Brown-Roberts-Wells
(BRW) head frame**

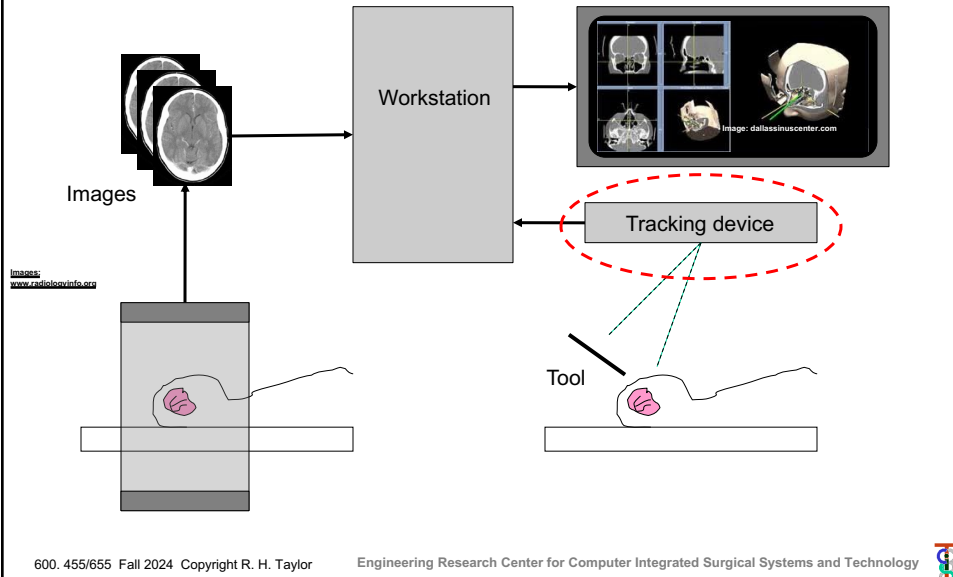
600. 455/655 Fall 2024 Copyright R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



2

Surgical Navigation Systems



3

3D localizers

- Determine 3D positions in space relative to some coordinate system
- Also called “3D digitizers”, “3D navigation systems”, “localizers”, etc.
- Many uses
- Many technologies

4

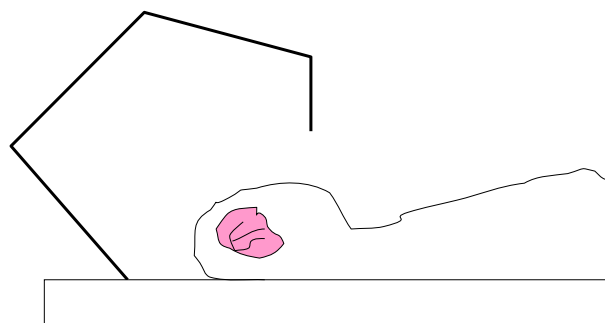
Localizer technologies

- Instrumented passive manipulator
- Active manipulator
- Ultrasound
- Electromagnetic
- Optical active
- Optical passive
- Miscellaneous – e.g., fiber optic



Passive mechanical linkages

- Encoders & linkage



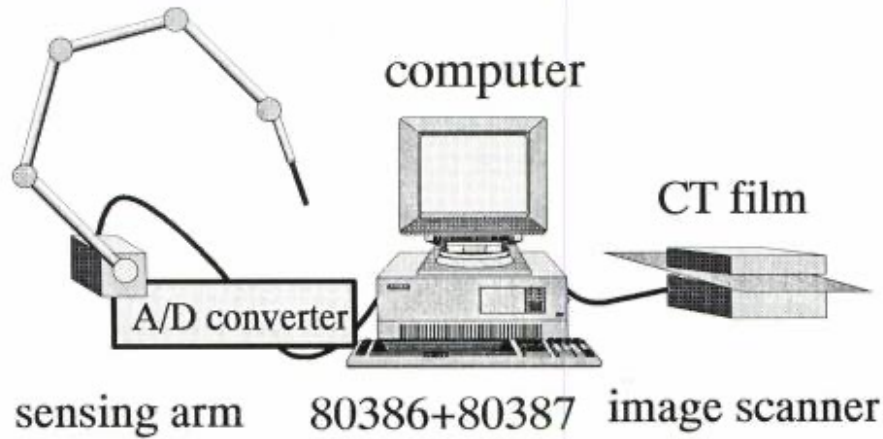


Figure 1. A schematic representation of the neuronavigator system. It consists of a microprocessor and a multi-articulated arm structure.

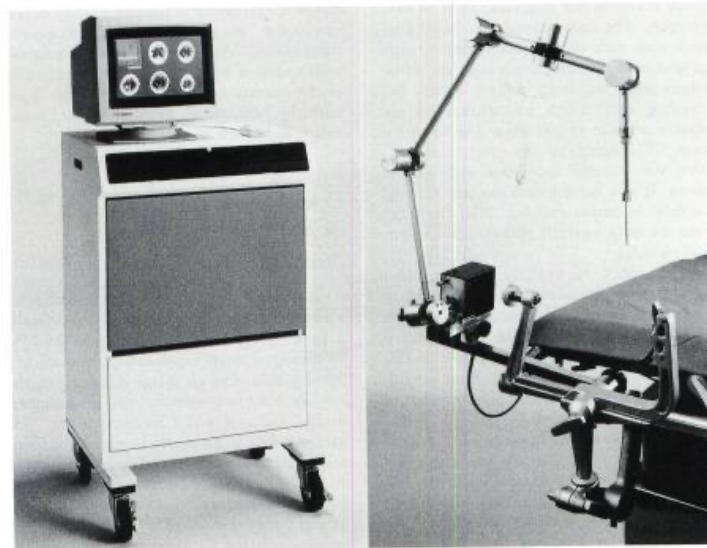
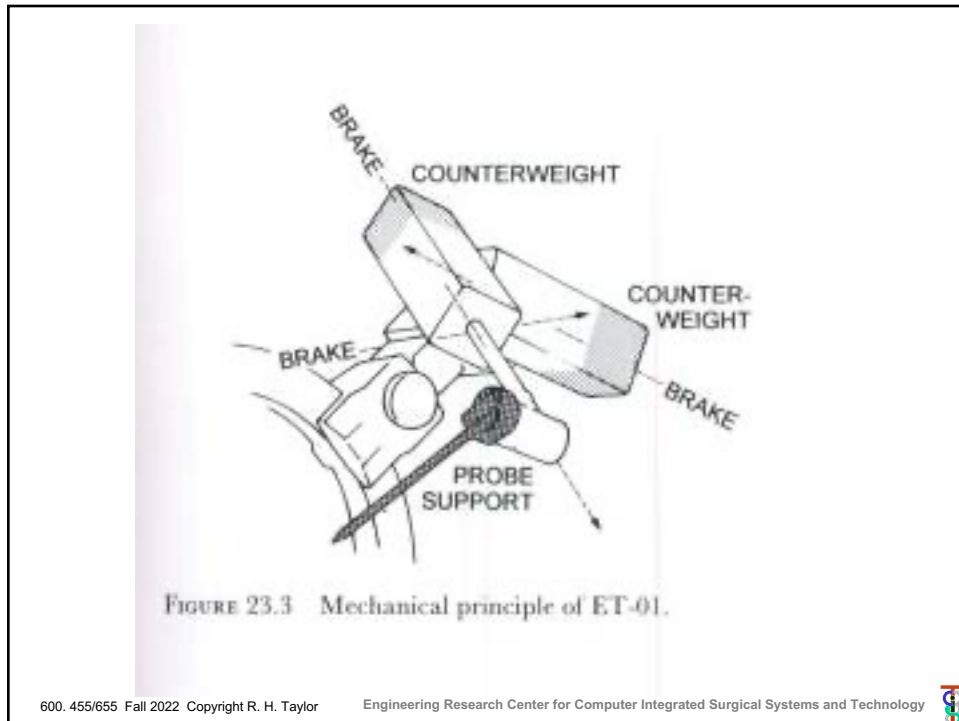
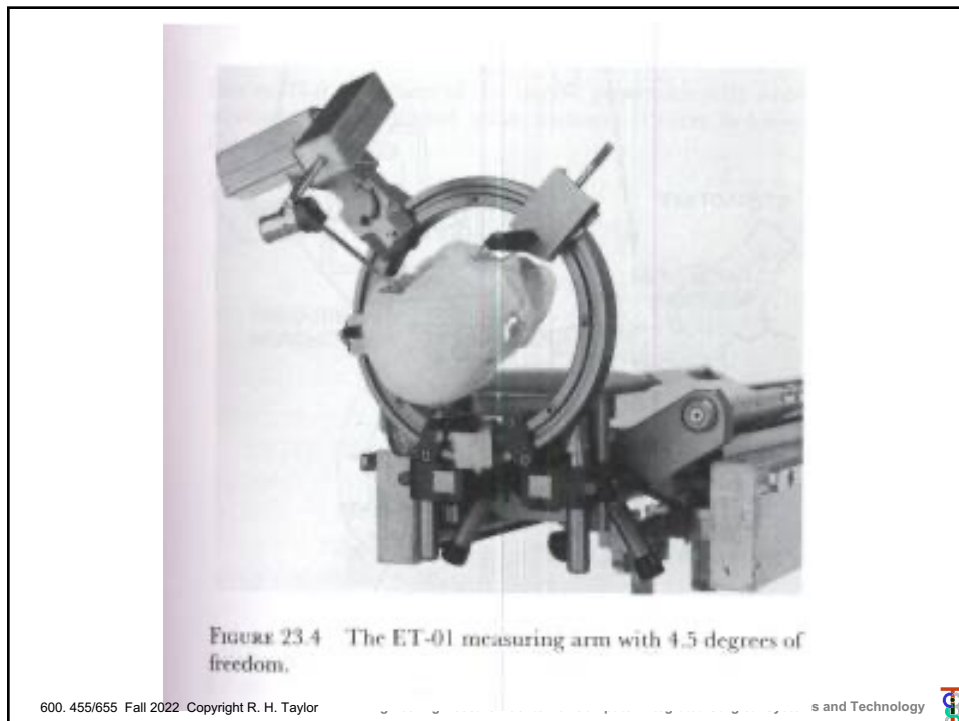


Figure 2. Photograph of the neuronavigator. The computer system is housed in the console box on the left. The sensing arm is secured to the Mayfield skull clamp. Six CT slices are displayed on the computer screen. Two cross markers display the location of the navigator tip.

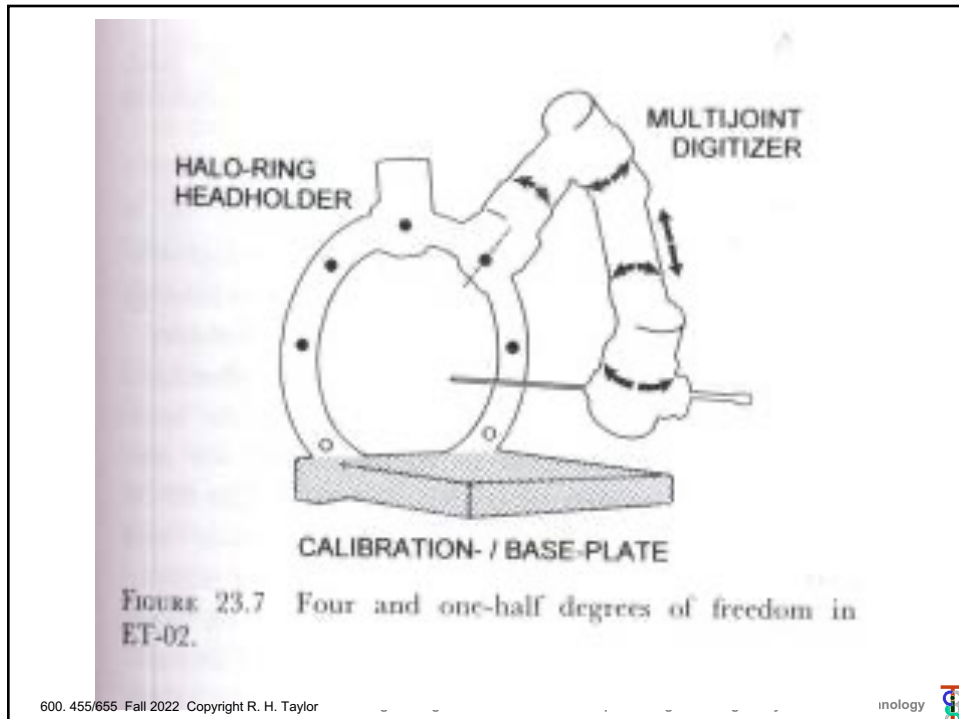




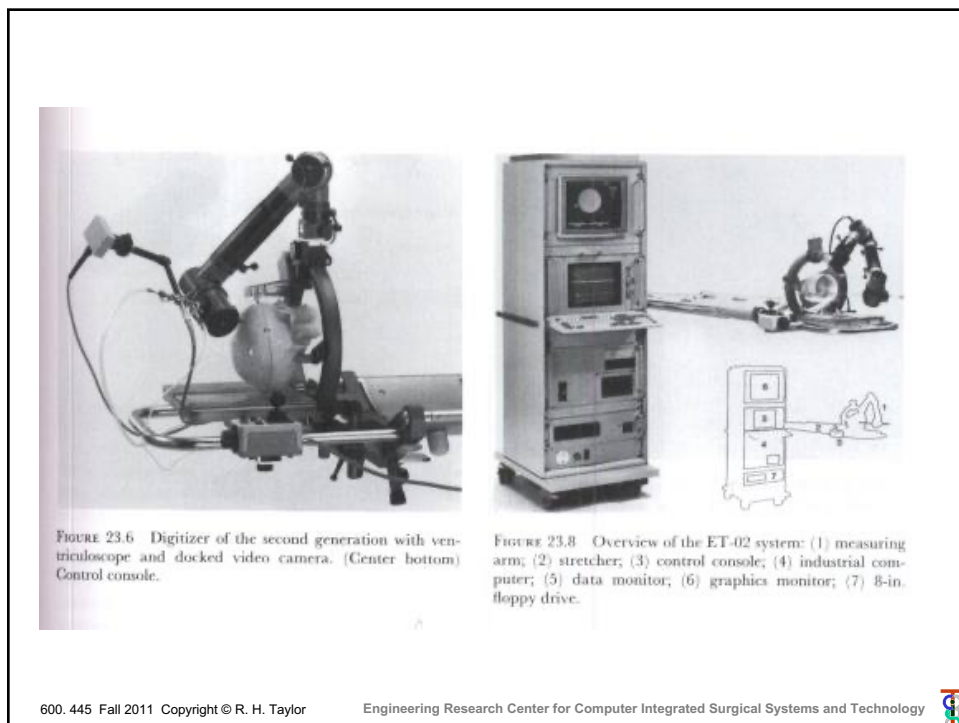
9



10



11



12

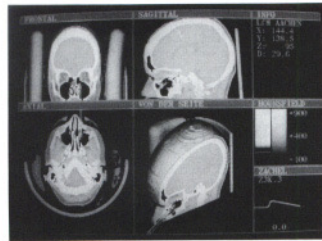


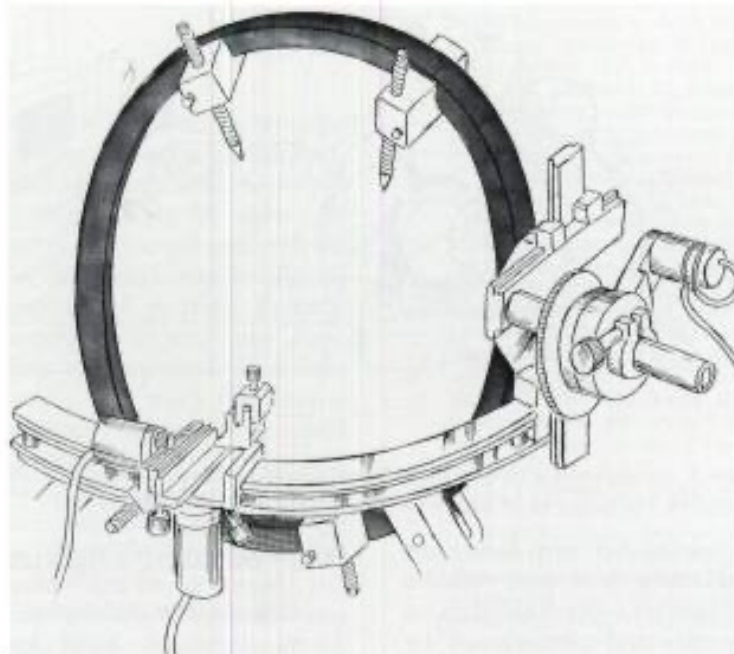
FIGURE 51.1 Image display on the CAS monitor screen.

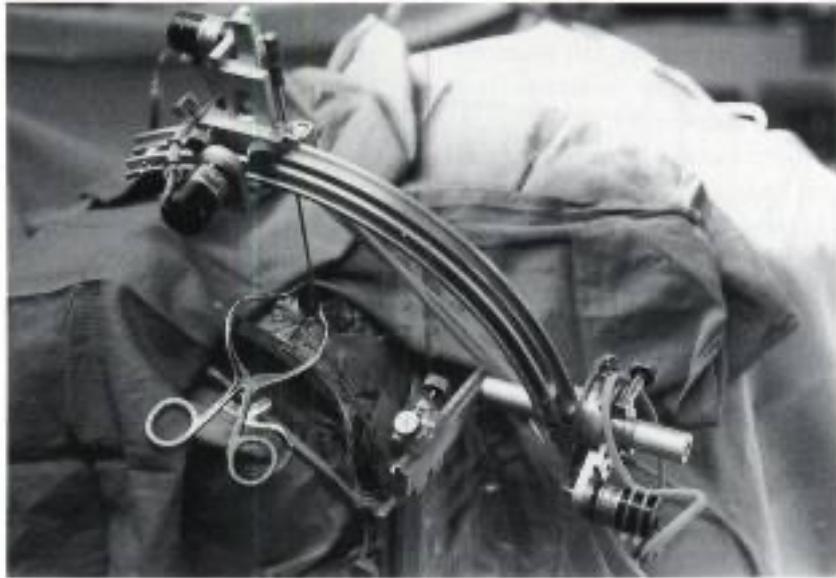
Therefore, we developed an appropriate measuring device which has 6 degrees of freedom [2, 3]. Digital increment encoders have been applied for shaft angle measurement. The pulse signals of the six rotary encoders are evaluated by 16-bit counters. A dedicated 68008 microcomputer calculates the position of the measuring probe from the measured angles and the given arm lengths. The system was developed with 3D imaging (figure 51.1).

A third generation of mechanical systems was developed to achieve better intraoperative handling [4] (figure 51.2). Counterbalanced arm elements allow for easy movements in every position. The 68008 was replaced by a PC-486.



FIGURE 51.2 The Aachen device for CAS with electro-mechanical measuring arm (coordinate digitizer).





600. 445 Fall 2011 Copyright © R. H. Taylor

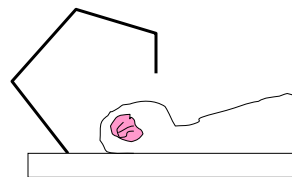
Engineering Research Center for Computer Integrated Surgical Systems and Technology



15

Passive mechanical linkages

- Encoders & linkage
- Advantages



600. 445 Fall 2011 Copyright © R. H. Taylor

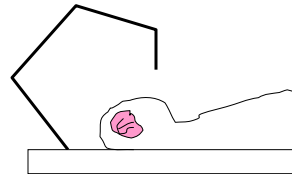
Engineering Research Center for Computer Integrated Surgical Systems and Technology



16

Passive mechanical linkages

- Encoders & linkage
- Advantages:
 - simple
 - no line-of-sight problems



600. 445 Fall 2011 Copyright © R. H. Taylor

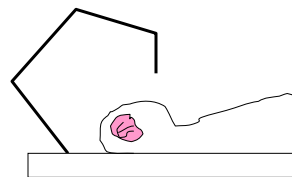
Engineering Research Center for Computer Integrated Surgical Systems and Technology



17

Passive mechanical linkages

- Encoders & linkage
- Advantages:
 - simple
 - no line-of-sight problems
- Drawbacks
 - clumsy
 - single frame
 - reference base



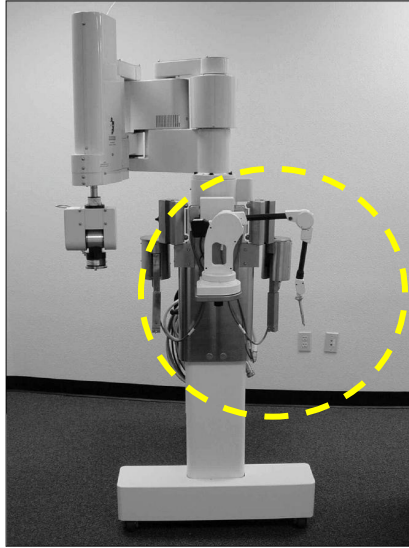
600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



18

Some commercially used examples



Neocis Yomi

Think Surgical Robodoc

600. 445 Fall 2011 Copyright © R. H. Taylor

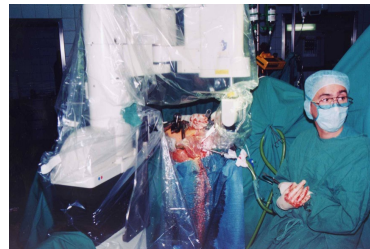
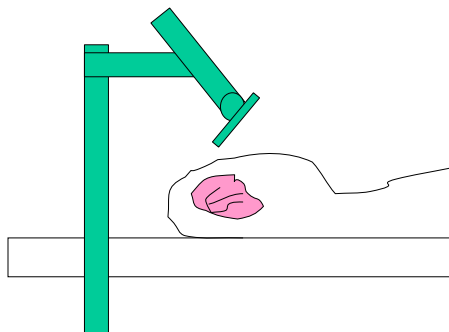
Engineering Research Center for Computer Integrated Surgical Systems and Technology



20

Active mechanical linkages

- Robot + hand guiding
- E.g., Robodoc, REMS



600. 445 Fall 2011 Copyright © R. H. Taylor

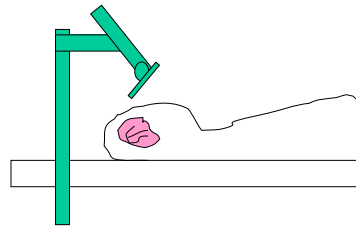
Engineering Research Center for Computer Integrated Surgical Systems and Technology



21

Active mechanical linkages

- Robot + hand guiding
- E.g., Robodoc
- Advantages



600. 445 Fall 2011 Copyright © R. H. Taylor

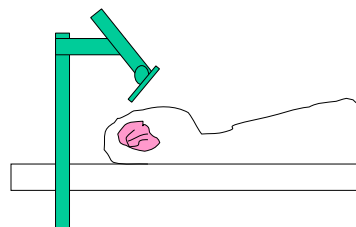
Engineering Research Center for Computer Integrated Surgical Systems and Technology



23

Active mechanical linkages

- Robot + hand guiding
- E.g., Robodoc
- Advantages
 - accurate
 - registered to robot
 - can combine with search, actions
- Drawbacks



600. 445 Fall 2011 Copyright © R. H. Taylor

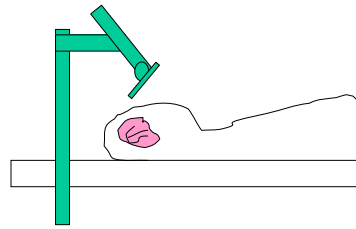
Engineering Research Center for Computer Integrated Surgical Systems and Technology



24

Active mechanical linkages

- Robot + hand guiding
- E.g., Robodoc
- Advantages
 - accurate
 - registered to robot
 - can combine with search, actions
- Drawbacks
 - clumsy
 - expensive
 - single tool, referencing



600. 445 Fall 2011 Copyright © R. H. Taylor

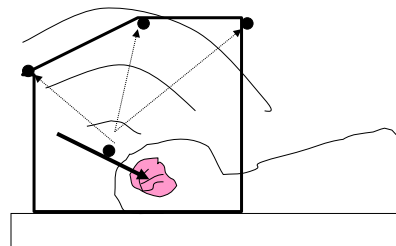
Engineering Research Center for Computer Integrated Surgical Systems and Technology



25

Ultrasound

- “Clickers”+microphones
- time delays give distances
- multiple distances give pos.



600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



26

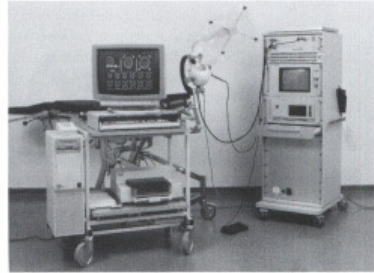


FIGURE 23.9 First sonic digitizer: system overview. Graphic computer (left), head-holder, sonic system (middle), SAC device, measuring computer (right).

intraoperative application. While we were evaluating the Science Accessories Corporation's (SAC) sonic system, we read about a first application of this device by Roberts [14] for the spatial, image-assisted localization of an operating microscope.

SONIC MICROSTEREOMETRY: ET-03

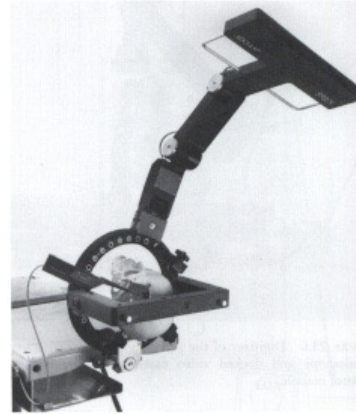


FIGURE 23.10 SPOCS with emitter panel (top), head-holder with detachable calibration frame, and supported targeting instrument (below left).



A Comparison of Sonic Digitizers Versus Light-Emitting Diode-Based Localization

189

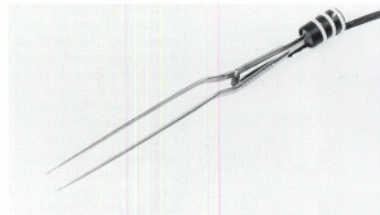


Figure 8. Sonic forceps, with wires directly attached to the emitters since the voltages involved precluded the use of a miniaturized connector.

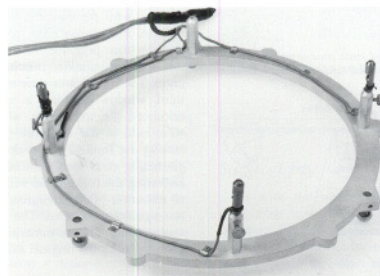


Figure 9. Sonic emitter ring, which attaches to the BRW head ring by the standard BRW ball/cam lock system; emitters can be placed at four points around the ring, allowing its use for posterior-based craniotomies.



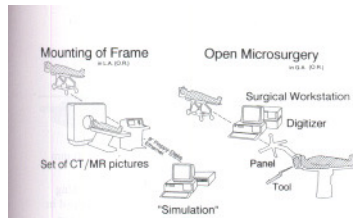


FIGURE 23.11 Sonic microstereometry. Data acquisition and operating procedure.

three microphones were docked with the ring at the same time, immediately after image data acquisition, or in a separate second procedure (see figure 23.11). The system was calibrated with a calibration panel containing three emitters, which were screwed temporarily to the ring. On the basis of the CT or MRI calibration marks, the software determines the exact position of each individual image relative to the calibration panel or base ring (first matrix operation) and then calculates the spatial relationship with the microphone panel (second matrix operation). The position of the targeting tool (one to four emitters) relative to the panel is determined finally in a third matrix operation.

The effect of interfering thermal factors could be largely eliminated by means of a measuring distance between the foot of the panel close to the head-retaining ring and the panel. Before each measuring cycle, a reference signal was emitted by the reference emitter and reached the panel microphones approximately 60 cm away in a known time of travel. Deviations (e.g., owing to temperature shifts) were taken into account auto-

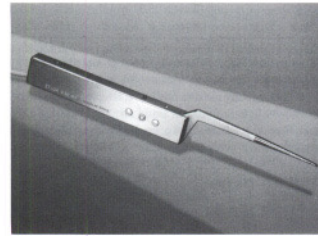


FIGURE 23.12 Bayonet-shaped standard measuring tool.

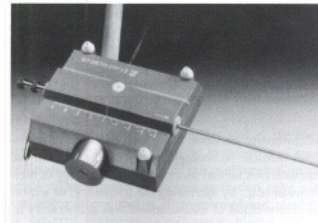
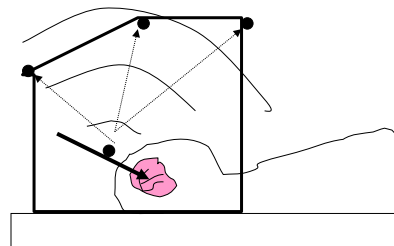


FIGURE 23.13 Measurement platform with four emitters for stereotaxy.



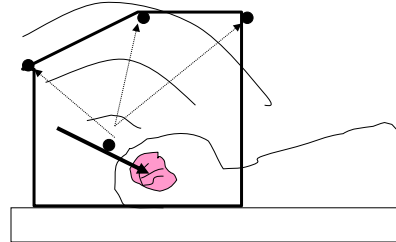
Ultrasound

- “Clickers”+microphones
- time delays give distances
- multiple distances give pos.
- Advantages



Ultrasound

- “Clickers”+microphones
- time delays give distances
- multiple distances give pos.
- Advantages
 - Cheap, unobtrusive
 - multiple rigid bodies



600. 445 Fall 2011 Copyright © R. H. Taylor

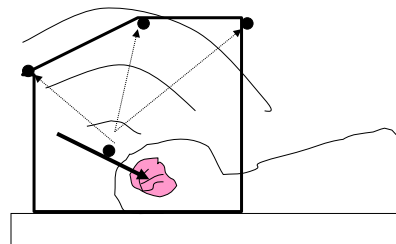
Engineering Research Center for Computer Integrated Surgical Systems and Technology



31

Ultrasound

- “Clickers”+microphones
- time delays give distances
- multiple distances give pos.
- Advantages
 - Cheap, unobtrusive
 - multiple rigid bodies
- Drawbacks
 - Accuracy drifts (e.g., temperature)
 - Lack of self-evident warning



600. 445 Fall 2011 Copyright © R. H. Taylor

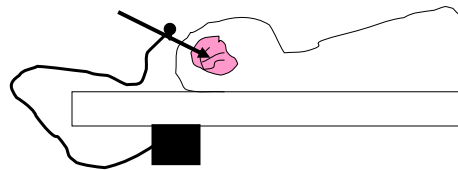
Engineering Research Center for Computer Integrated Surgical Systems and Technology



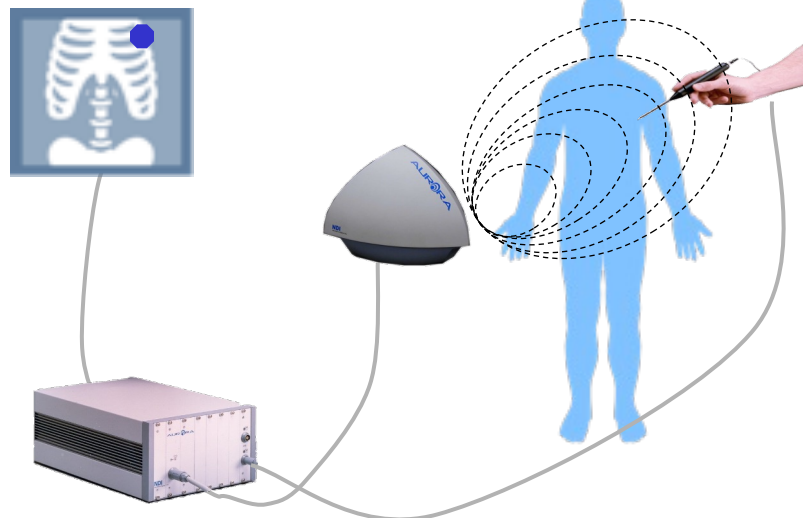
32

Electromagnetic

- Originally developed for fighter pilot head tracking
- Reasonably accurate 6 dof
- E.g., Polhemus, Ascension, NDI Aurora
- Advantages
- Drawbacks



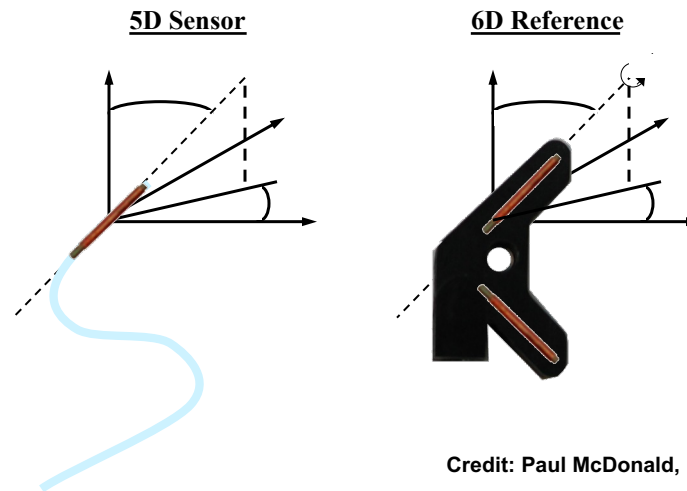
How Does An EM System Work?



Credit: Paul McDonald, NDI



How a Magnetic System Works



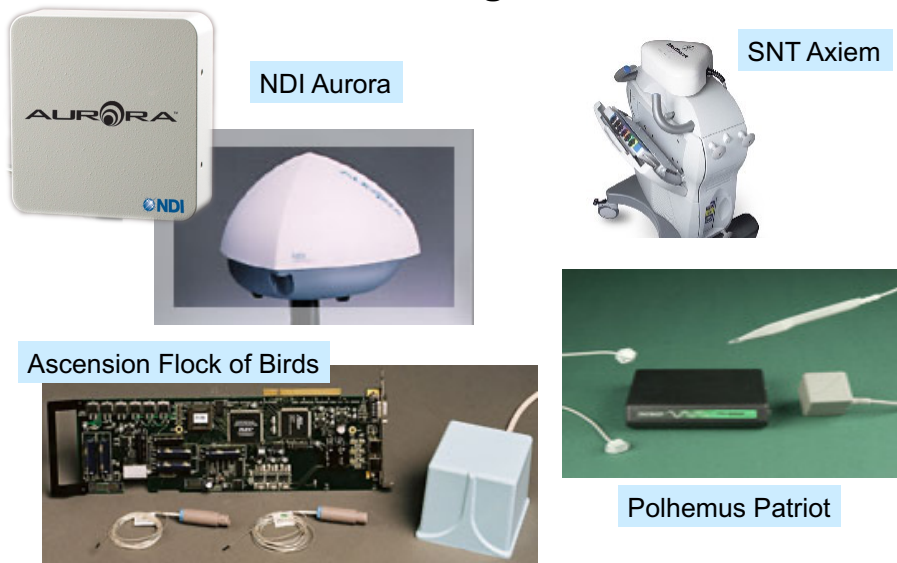
600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



36

Current Electromagnetic Products



600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



37

Example: NDI Aurora™

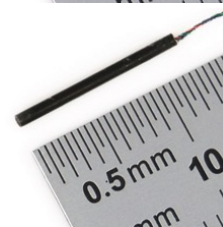
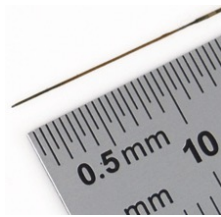
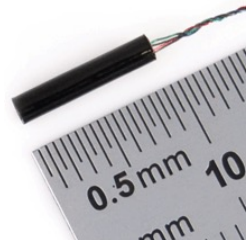
Transmitter



5 DOF Sensors



6 DOF Sensors



<http://www.ndigital.com/medical/aurora-techspecs.php>

600. 445 Fall 2011 Copyright © R. H. Taylor

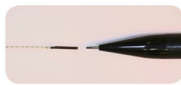
Engineering Research Center for Computer Integrated Surgical Systems and Technology



38

Example: ATI medSAFE™ (Sensors)

Sensor options:

Model 90 6DOF Sensor		Model 90 6DOF Sensor • Sensor OD = 0.9 mm • Sensor Length = 7.25 mm • Cable OD = 0.6 mm • Cable length = 6.6 ft. (2.0 m)
Model 130 6DOF Sensor		Model 130 6DOF Sensor • Sensor OD = 1.5 mm • Sensor Length = 7.6 mm • Cable OD = 1.2 mm • Cable length = 6.6 ft. (2.0 m)
Model 180 6DOF Sensor		Model 180 6DOF Sensor • Sensor OD = 2 mm • Sensor Length = 9.9 mm • Cable OD = 1.2 mm • Cable length = 6.6 ft. (2.0 m)
Model 800 6DOF Sensor		Model 800 6DOF Sensor • Sensor OD = 7.9 mm • Sensor Length = 19.8 mm • Cable OD = 3.8 mm • Cable length = 6.6 ft. (2.0 m)

<http://www.ascension-tech.com/medical/medSAFE.php>

600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



39

Example: ATI medSAFE™ (Transmitters)

Transmitter options:

Short-Range Transmitter		Set-Up & Use: 6.3 cm (2.5 inches) x 4.6 cm (1.8 inches) x 5.2 cm (2.1 inches). Transmitter weighs just 290 grams. It generates pulsed DC magnetic fields for high accuracy tracking in short-range applications. Ranges: • 46 cm for Model 800 Sensor; contact Ascension for performance of smaller sensors with this transmitter.
Mid-Range Transmitter		Set-Up & Use: 9.6 cm (3.8 inches) cube that generates pulsed DC magnetic fields for high accuracy tracking over medium ranges. Ranges: • 36 cm for Model 90 Sensor • 46 cm for Model 130 Sensor • 58 cm for Model 180 Sensor • 78 cm for Model 800 Sensor
Flat Transmitter		Set-Up & Use: 56 cm (22 inches) x 56 cm (22 inches) x 2.54 cm (1 inch). Flat transmitter is for unobtrusive placement beneath a patient. It generates a field above planar surface while negating any possible distortion of measurements by ferrous metal in an OR procedural table. Ranges: • 40 cm for Model 90 Sensor • 46 cm for Model 130 Sensor • 46 cm for Model 180 Sensor • 46 cm for Model 800 Sensor

<http://www.ascension-tech.com/medical/medSAFE.php>

600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



40

Electromagnetic

Pros

- No line of sight required
- Tools can be populated with small sensors
- Generally less expensive than optical

Cons

- Metal Interference
- Less stable than optical
- Smaller measurement volume
- Incapable of tracking more than a few 6DOF tools

Credit: Paul McDonald, NDI

600. 445 Fall 2011 Copyright © R. H. Taylor

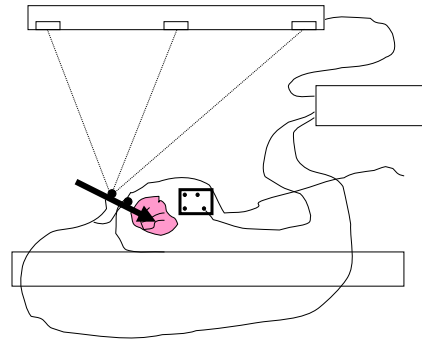
Engineering Research Center for Computer Integrated Surgical Systems and Technology



42

Optical active

- Track LED markers
- Triangulate to locate 3D
- E.g.: Optotrak, Pixsys
- Current “gold standard”
- Advantages
- Disadvantages



A Comparison of Sonic Digitizers Versus Light-Emitting Diode-Based Localization

191

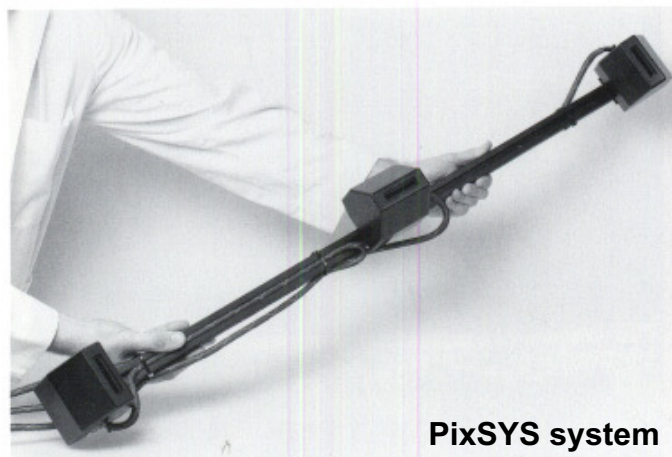


Figure 12. The optical camera system as implemented in the authors' system mounted on an aluminum extrusion.



Optical Active



NDI Optotrak



NDI Optotrak



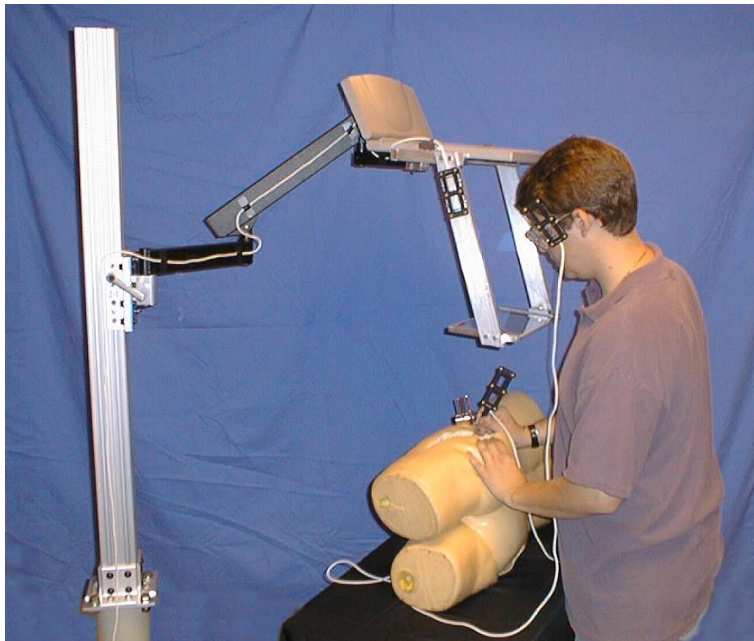
Flashpoint

600. 455/655 Fall 2022 Copyright R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



45



600. 455/655 Fall 2022 Copyright R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



46

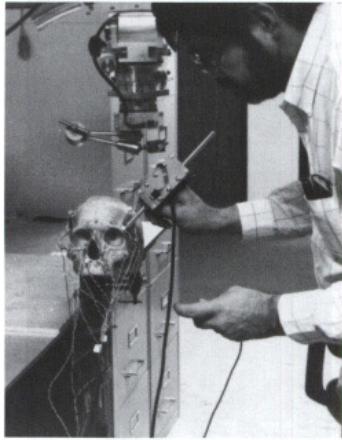
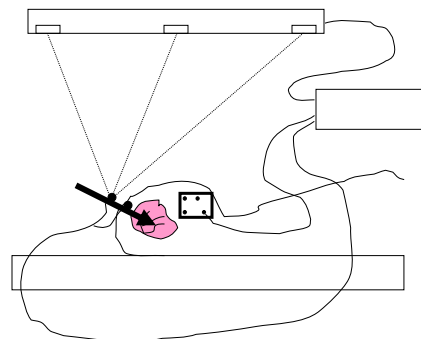


FIGURE 52.8 Locating landmarks on the skull. The position of the tip of the digitizer wand relative to the beacons on the wand has previously been calibrated. The positions of the beacons mounted to the skull are continuously monitored to provide a base coordinate system for the landmark location. Once the skull has been located, the positions of the beacons relative to the preoperative skull coordinate system may be computed.



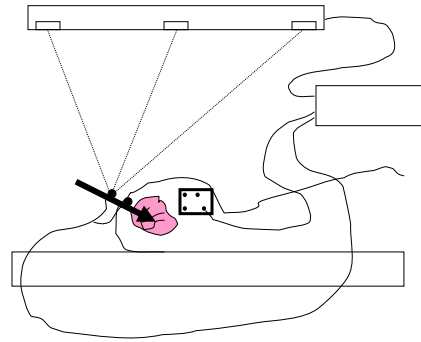
Optical active

- Track LED markers
- Triangulate to locate 3D
- E.g.: Optotrak, Pixsys
- Current “gold standard”
- Advantages



Optical active

- Track LED markers
- Triangulate to locate 3D
- E.g.: Optotrak, Pixsys
- Current “gold standard”
- Advantages
 - very accurate
 - multiple rigid bodies
 - versatile
 - reasonably fail-safe
- Disadvantages



600. 455/655 Fall 2022 Copyright R. H. Taylor

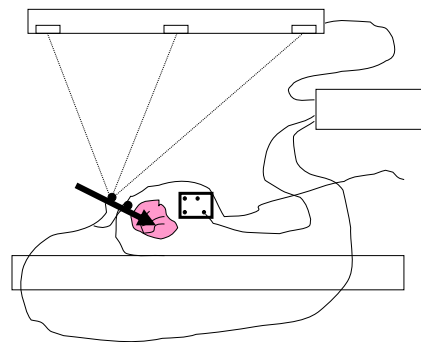
Engineering Research Center for Computer Integrated Surgical Systems and Technology



49

Optical active

- Track LED markers
- Triangulate to locate 3D
- E.g.: Optotrak, Pixsys
- Current “gold standard”
- Advantages
 - very accurate
 - multiple rigid bodies
 - versatile
 - reasonably fail-safe
- Disadvantages
 - line-of-sight restrictions
 - large, expensive



600. 445 Fall 2011 Copyright © R. H. Taylor

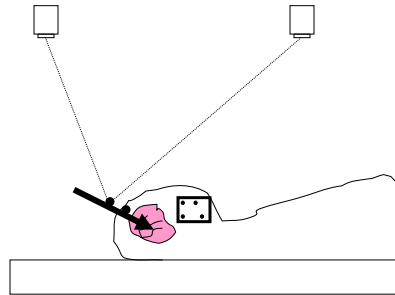
Engineering Research Center for Computer Integrated Surgical Systems and Technology



50

Optical passive

- Triangulate markers in standard video images or specialized IR cameras
- E.g.,
 - Heilbrun, Colchester, Mathelin, ...
 - Polaris
 - Atracsys
 - Claron



51

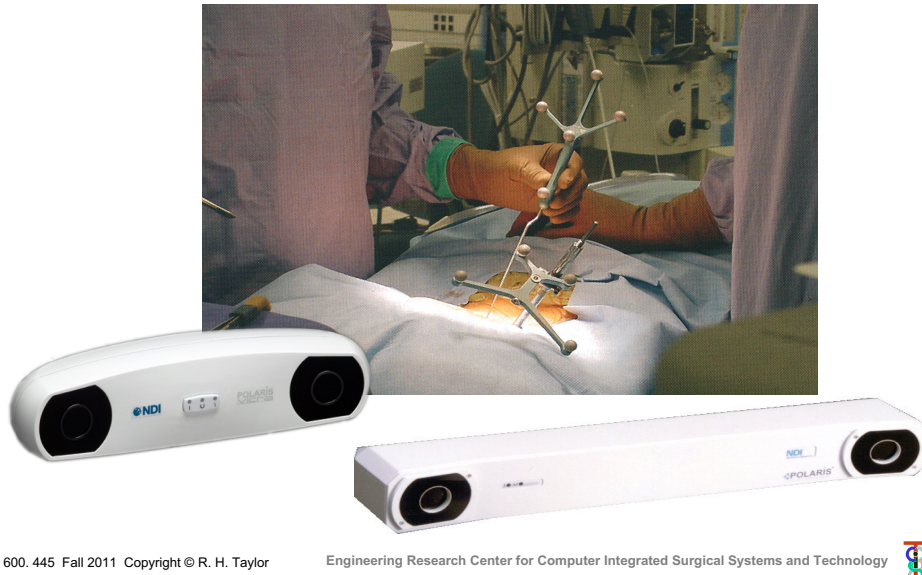


FIGURE 51.3 Optical position measurement for CAS.



52

Commercial Example (Reflective Markers): NDI Polaris



600. 445 Fall 2011 Copyright © R. H. Taylor

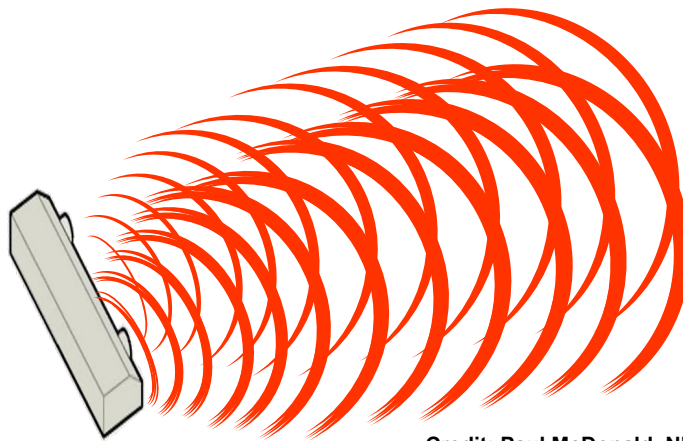
Engineering Research Center for Computer Integrated Surgical Systems and Technology



53

How does the Polaris system work?

The illuminators flood the area with infrared light



Credit: Paul McDonald, NDI

600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology

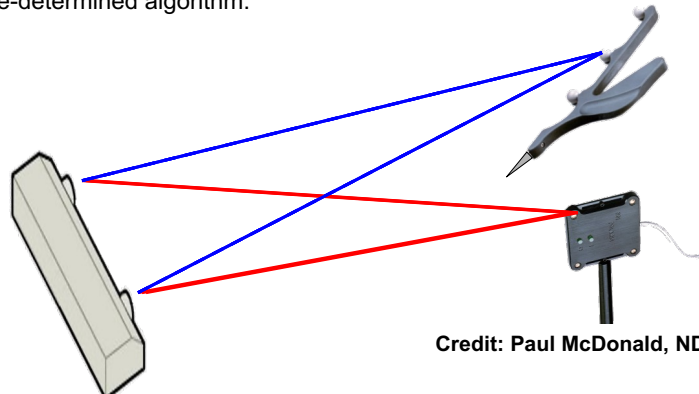


54

How does the Polaris system work?

The infrared light is **reflected** back to the Position Sensor by the passive markers, while active markers **emit** infrared light.

By calculating the position of each individual marker on a tool, the System is able to determine the exact location of the tip of the tool using a pre-determined algorithm.



Credit: Paul McDonald, NDI

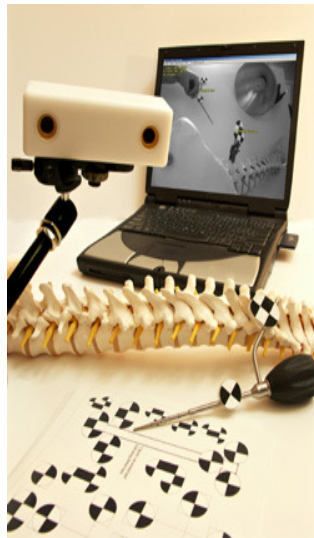
600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



55

Commercial Example (Ordinary Video): Claron Technology



<http://www.clarontech.com>



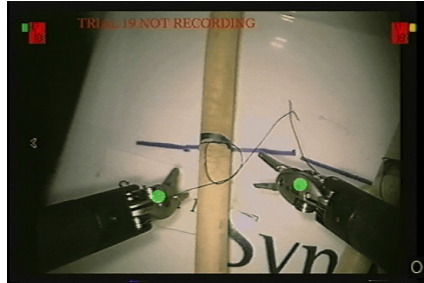
600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology

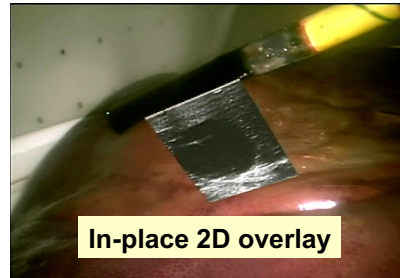
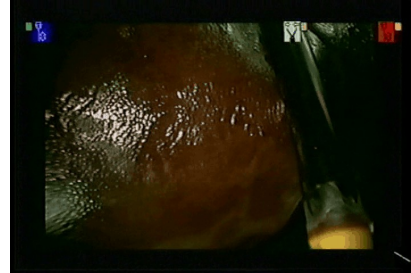


56

JHU research examples: tool tracking



Track video of tools in
mono or stereo images



600. 445 Fall 2011 Copyright © R. H. Taylor

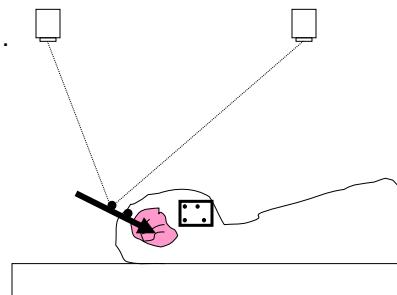
Engineering Research Center for Computer Integrated Surgical Systems and Technology



57

Optical passive

- Triangulate markers in standard video images or specialized IR cameras
- E.g.,
 - Heilbrun, Colchester, Mathelin, ...
 - Polaris, Claron
- Advantages
 - Inherent alignment for overlay
 - Same method thru microscope
 - Standard components
 - Fairly fail-safe
- Drawbacks
 - More computing needed (but special hardware possible)
 - Line-of-sight
 - Video resolution



600. 445 Fall 2011 Copyright © R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



58

Stereo Optical Summary

Pros

- Industry Standard
- Well known and defined performance characteristics
- Ability to track large multiple of tools simultaneously
- Accuracy typically below 0.35 mm RMS
- Large measurement volume
- Variety of targets can be affixed to the tool (IRED, sphere)
- Video self alignment [rht]

Cons

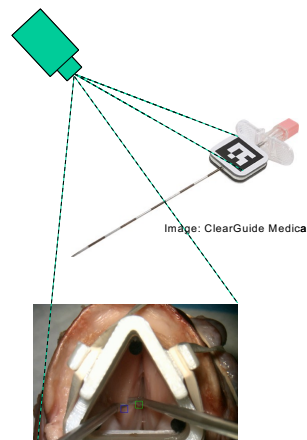
- Line-of-sight required
- Extraneous IR (sunlight)
- Rigid body tracking is most accurate, unable to track flexible devices
- Historically more costly when compared to other technologies
- Larger tools

Credit: Paul McDonald, NDI



Monocular Vision Tracking

- Use computer vision methods to locate surgical tools or specialized markers from conventional monocular video
- Frequently used in mixed reality applications
- Advantages
- Disadvantages

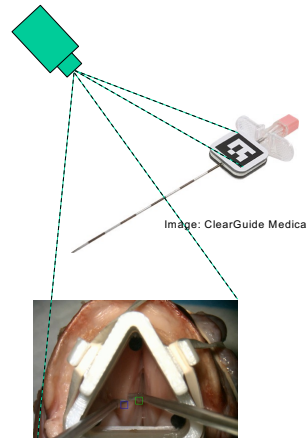


Sue Min Cho*, Henry H. Joo*, *et al.*
Otolaryngology-Head and Neck Surgery,
2023



Monocular Vision Tracking

- Use computer vision methods to locate surgical tools or specialized markers from conventional monocular video
- Frequently used in mixed reality applications
- Advantages
 - Cheap
 - Markers easy to fabricate
 - Can be markerless if tool is has sufficient visual landmarks
- Disadvantages



Sue Min Cho*, Henry H. Joo*, *et al.*
Otolaryngology-Head and Neck Surgery,
2023

600. 455/655 Fall 2024 Copyright R. H. Taylor

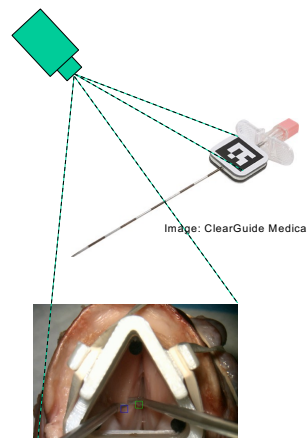
Engineering Research Center for Computer Integrated Surgical Systems and Technology



61

Monocular Vision Tracking

- Use computer vision methods to locate surgical tools or specialized markers from conventional monocular video
- Frequently used in mixed reality applications
- Advantages
 - Cheap
 - Markers easy to fabricate
 - Can be markerless if tool is has sufficient visual landmarks
- Disadvantages
 - Markers can take up large portion of visual field
 - Tools can be obscured
 - Less accurate (especially in depth)



Sue Min Cho*, Henry H. Joo*, *et al.*
Otolaryngology-Head and Neck Surgery,
2023

600. 455/655 Fall 2024 Copyright R. H. Taylor

Engineering Research Center for Computer Integrated Surgical Systems and Technology



62